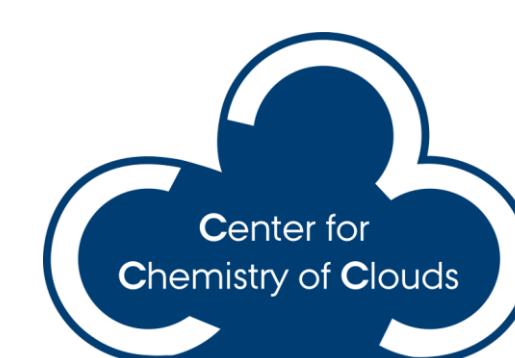


Direct Measurements of Oxygen in Organic Aerosol Particles and Possible Implications on Atmospheric Chemistry

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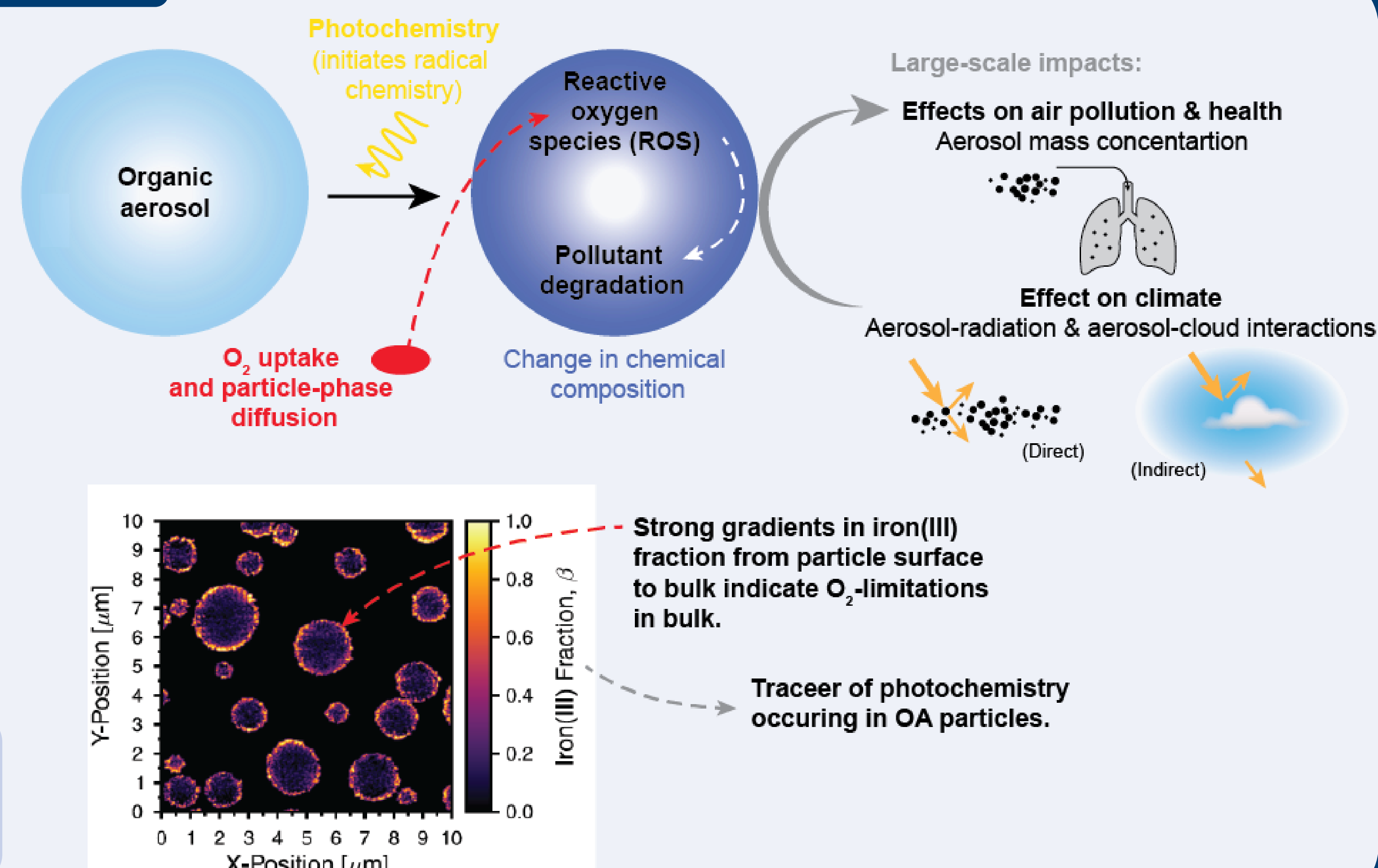


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Motivation

- Organic aerosols (OAs) make up **20–70% of atmospheric particulate matter**, and impacts **air quality, climate, and human health**[1].
- Oxygen (O₂) drives radical reactions** within OAs, shaping the composition and persistence of harmful compounds like reactive oxygen species (ROS) and polyaromatic hydrocarbons (PAHs) [2,3].
- Limited O₂ availability **slows radical chemistry**, allowing pollutants to live longer and altering overall aerosol reactivity[4].
- O₂ diffusion rates within particles are **strongly dependent on OA phase state**, which shifts with relative humidity and temperature across a spectrum from **liquid-like to amorphous solid states**[5,6].
- Highly viscous OAs are expected to severely **hinder O₂ diffusion**, yet **direct in-particle O₂ measurements are still lacking**, leaving impacts on aerosol reactivity and large scale atmospheric chemistry poorly constrained.

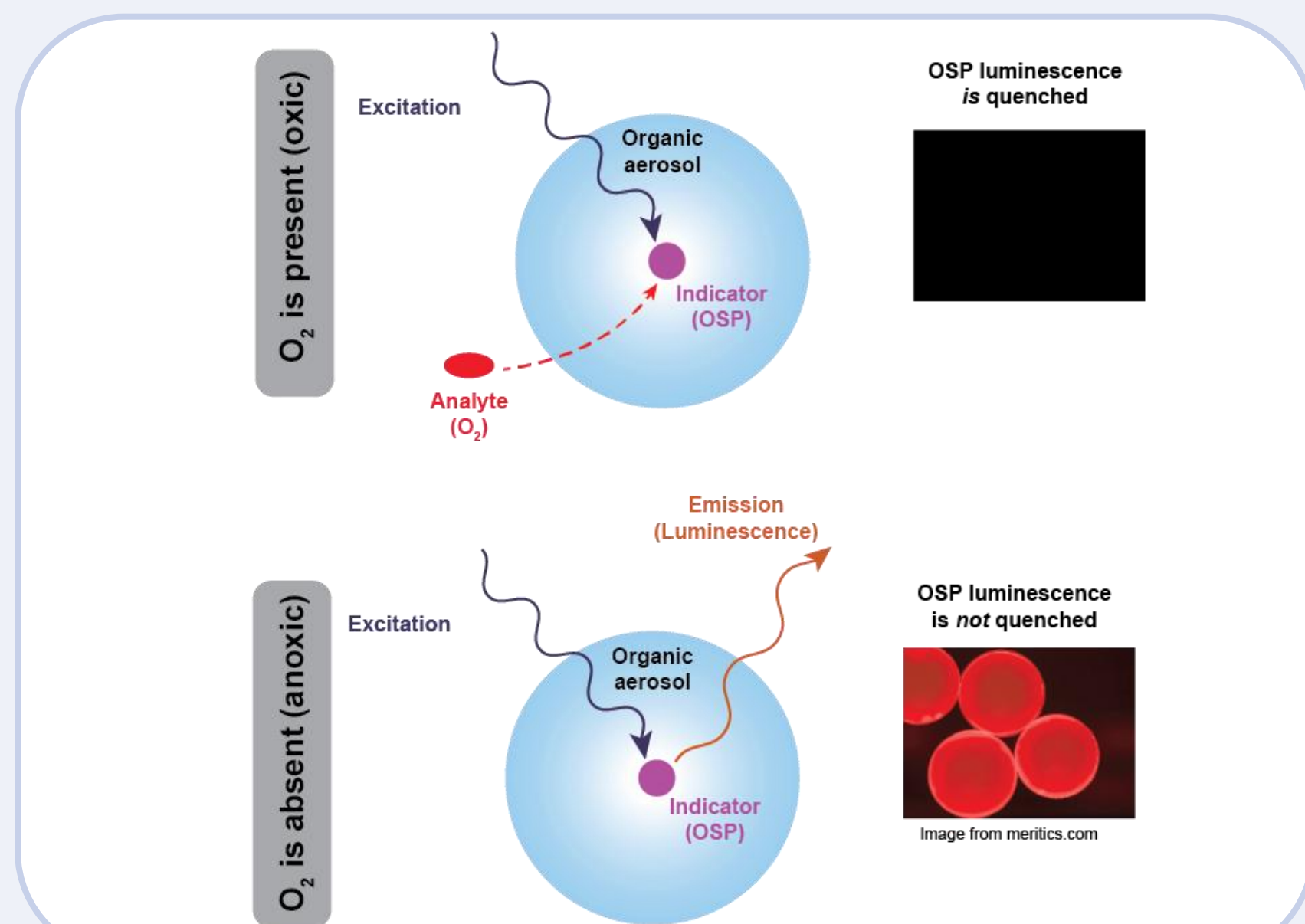
Objective: Develop a method to directly image and quantify O₂ availability inside individual organic aerosol particles, and probe how particle phase state controls O₂ diffusion and availability within particles.



Method

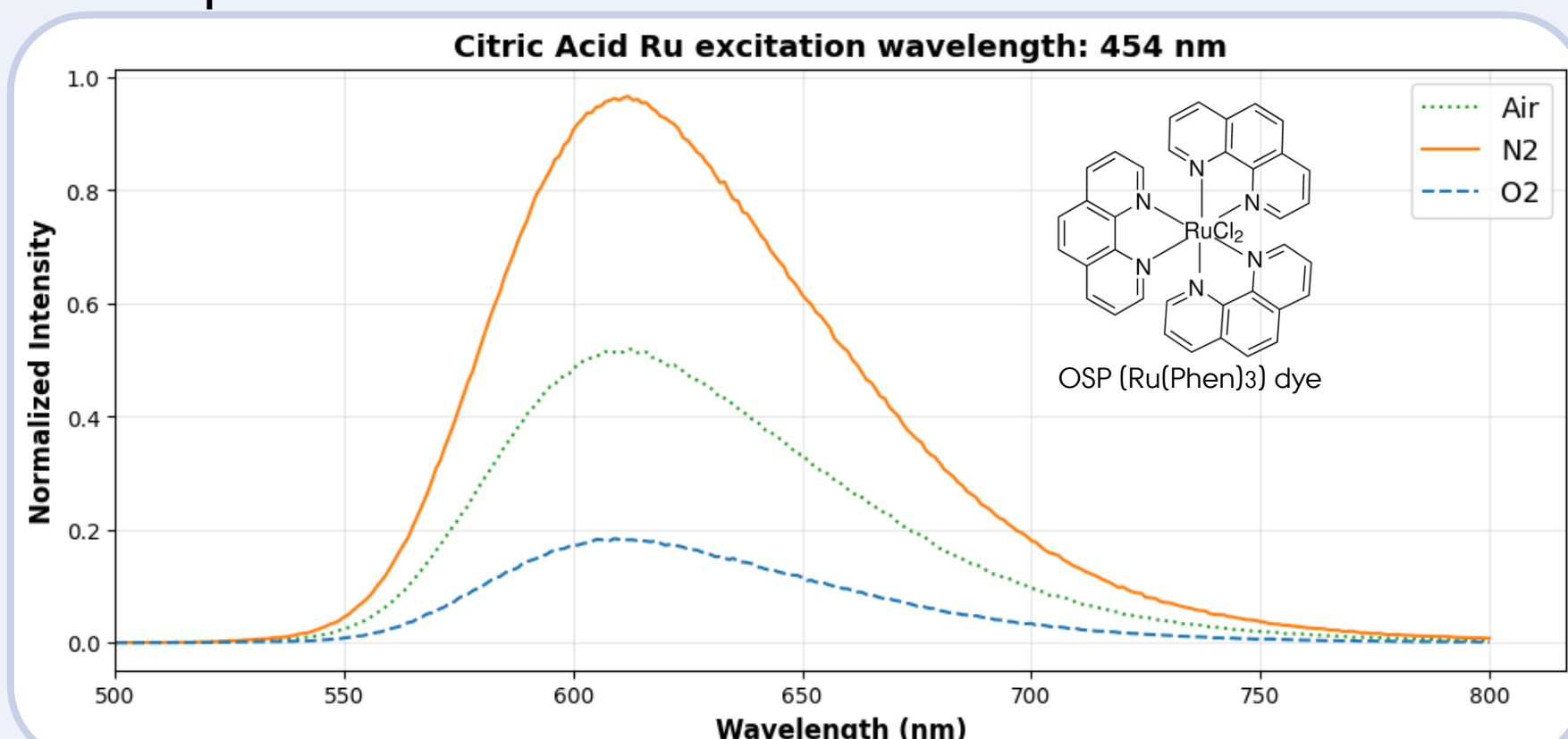
① The Optical Probe - OSP

Oxygen-sensitive probes (OSP) are embedded into particles of OA proxies (e.g. citric acid). Luminescence is **quenched by O₂**, providing a direct spatially resolved readout of O₂ availability.



Method Validation using Bulk Solutions:

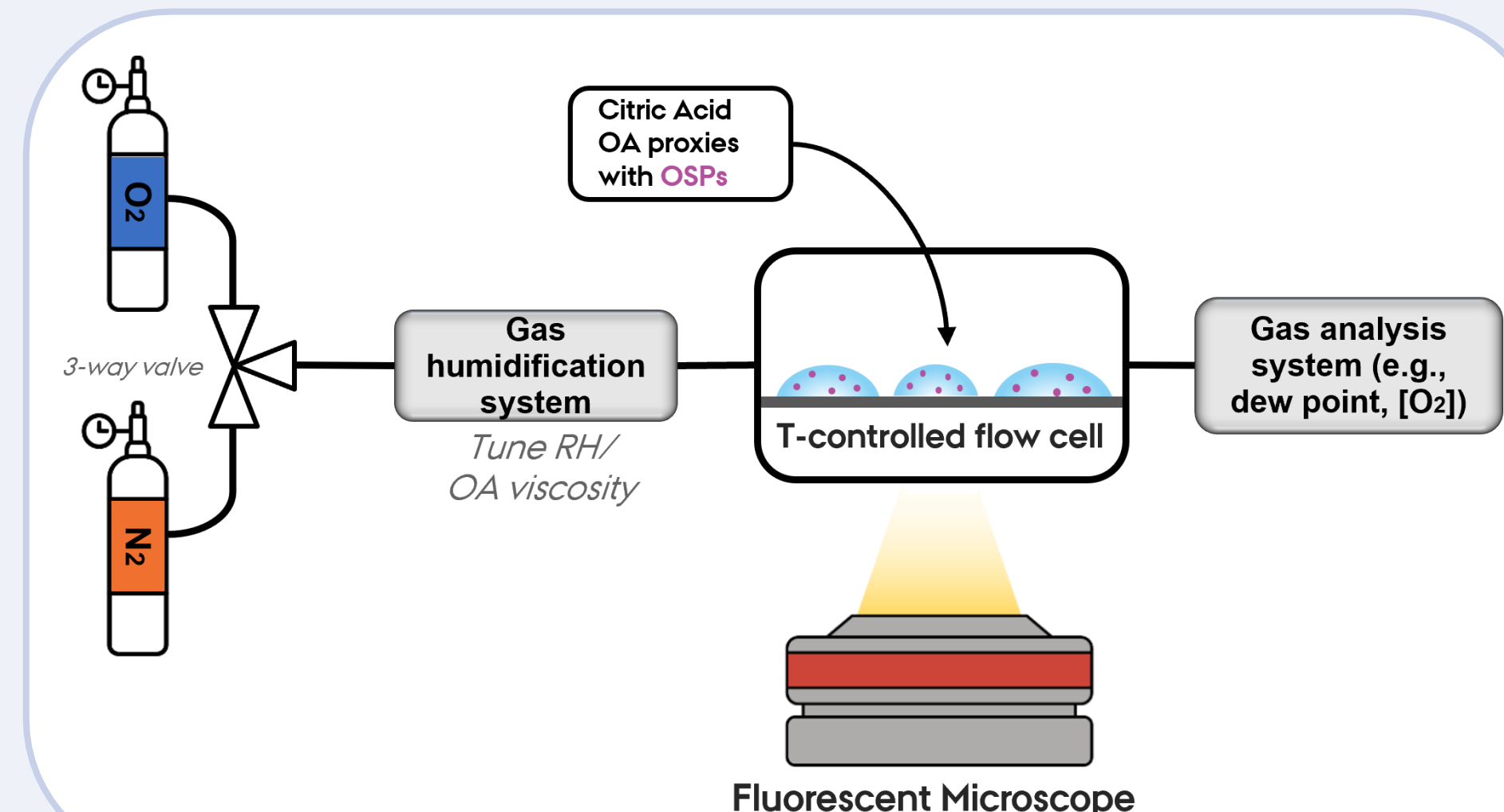
OSP fluorescence responds sensitively to O₂ in citric acid solution across N₂, air, and O₂ atmospheres.



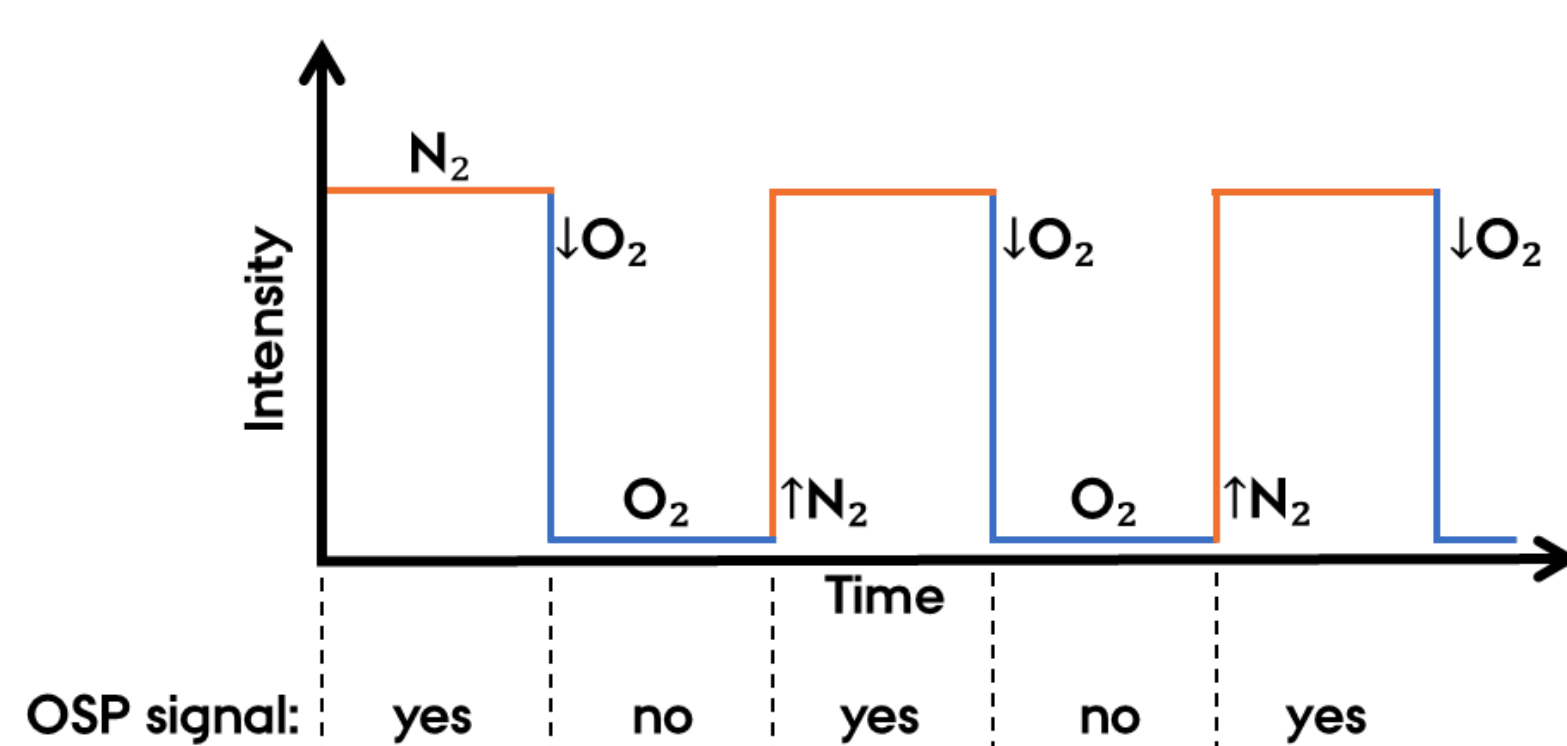
② Microscope Setup – Environmental Flow Cell

Particles are deposited in a **temperature-controlled flow cell** connected to a gas humidification system. O₂/N₂ mixtures of known composition are flushed through, enabling systematic control of:

- O₂ concentration** – oxic to anoxic conditions
- Relative humidity (RH)** – tuning particle viscosity
- Temperature** – extending phase state range

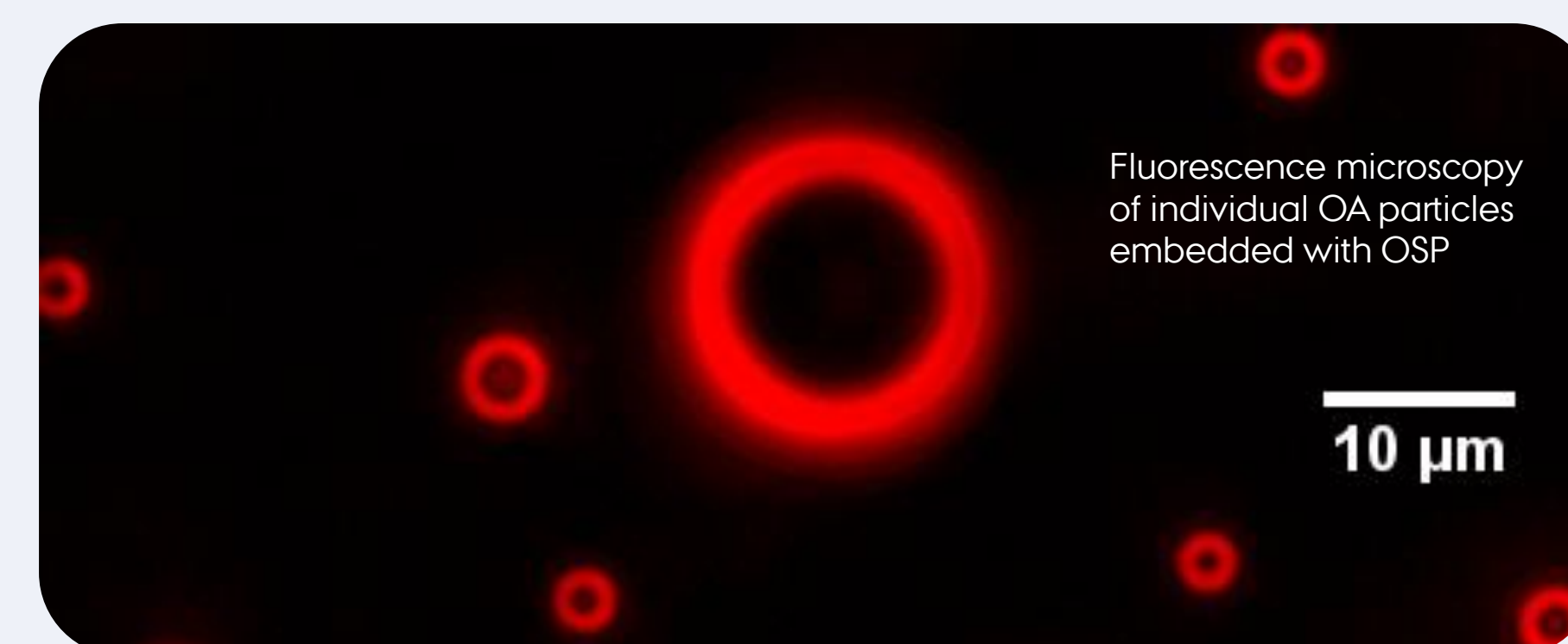


OSP Signal Reversible Switch

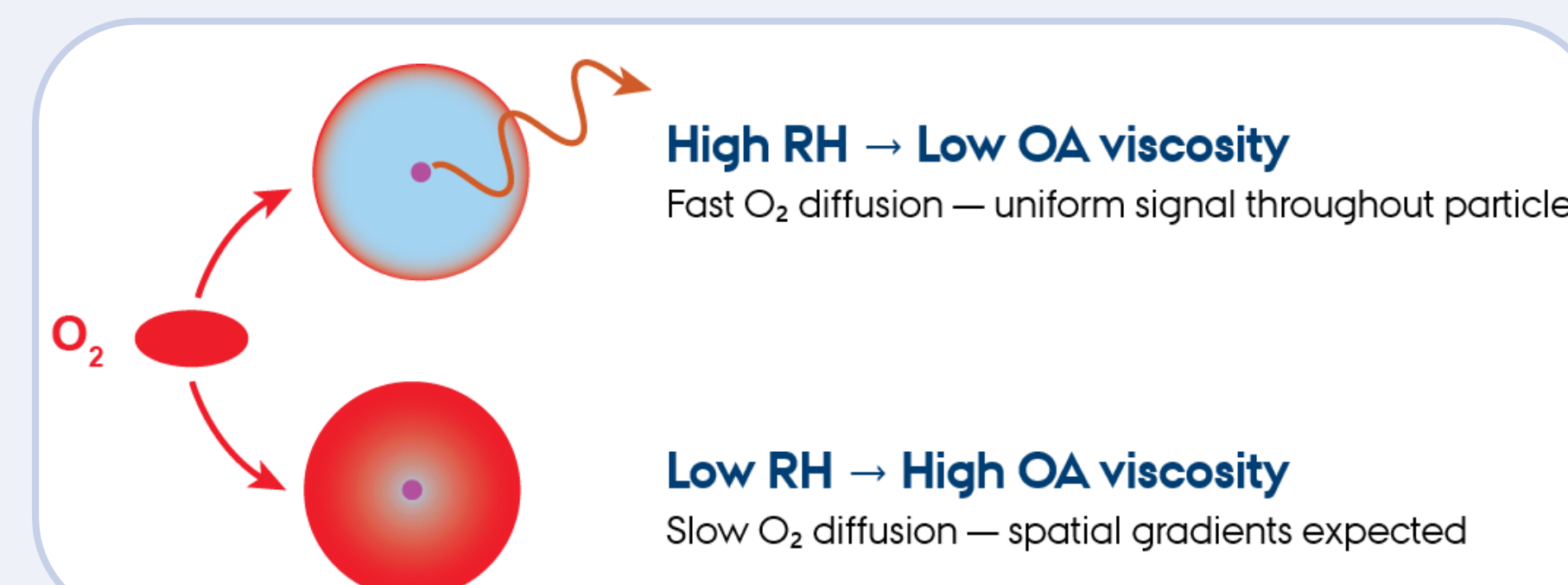


③ Imaging & What We Measure

Fluorescent microscopy resolves the spatial OSP signal within individual particles, enabling direct visualisation of oxic and anoxic conditions.



RH-Dependent Phase Effects on O₂ Availability



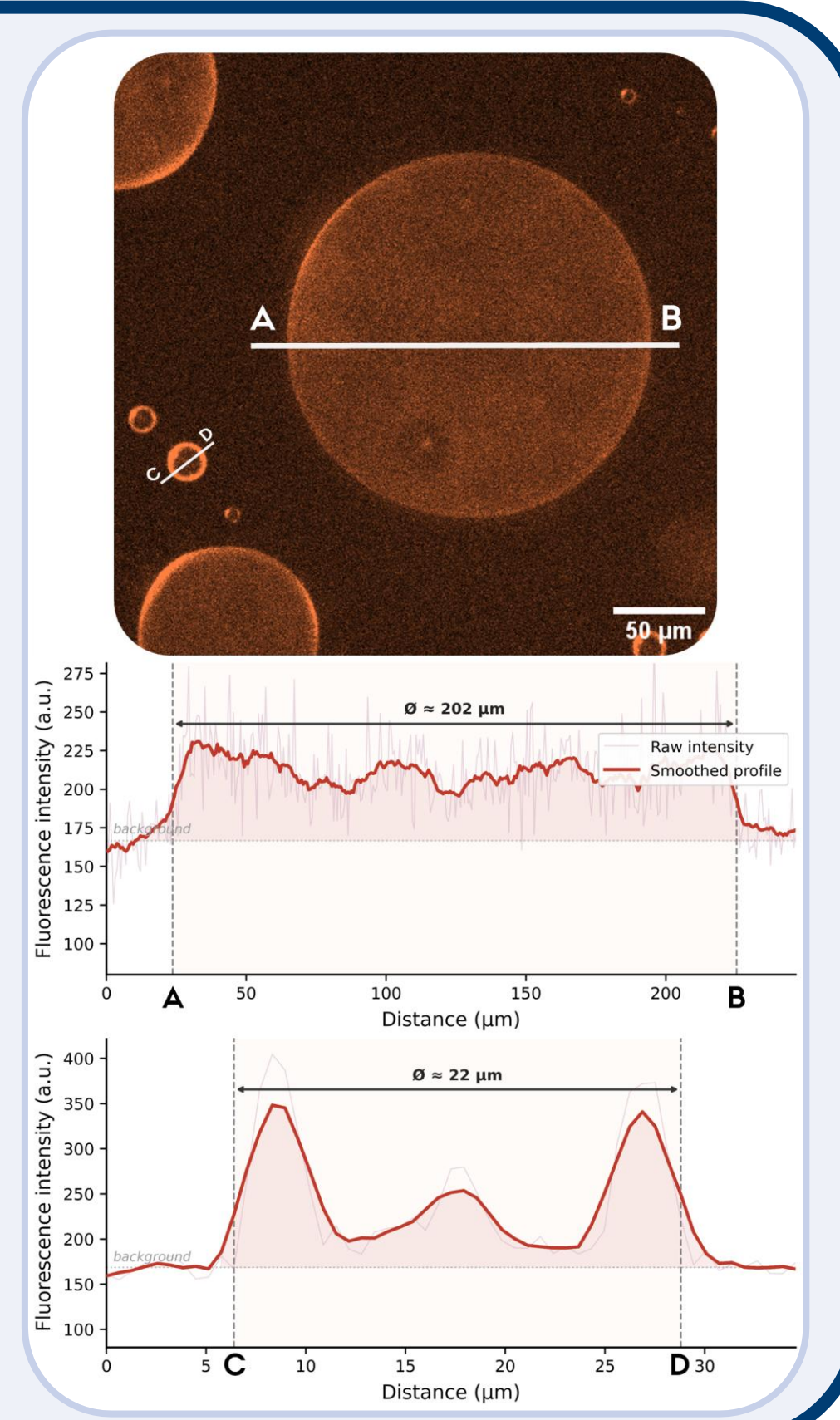
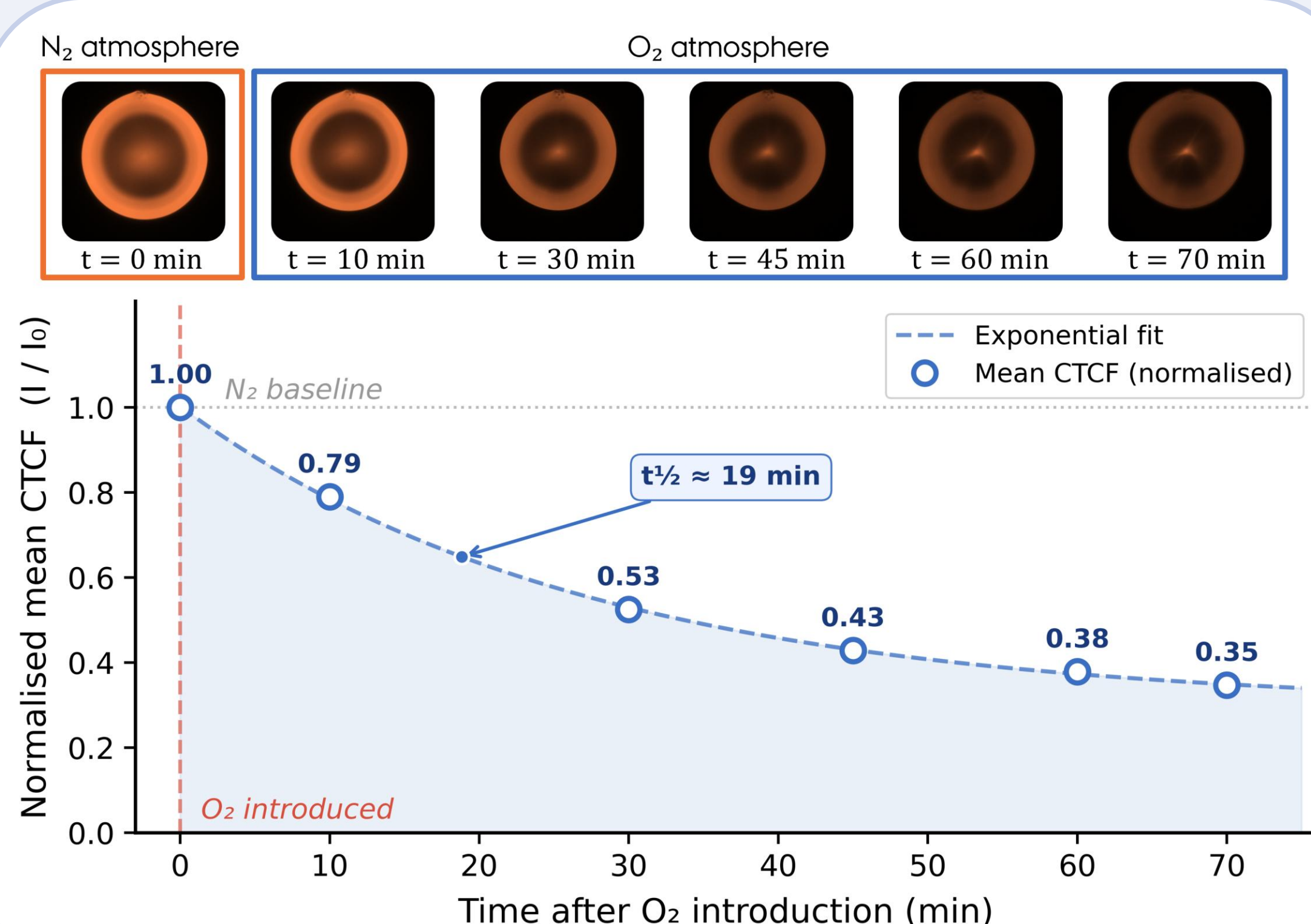
Quantitative O₂ Mapping

$$\tau_0/\tau = 1 + K_{SV} [O_2]$$

FLIM converts fluorescence lifetime into [O₂], enabling full quantitative mapping of O₂ gradients inside single particles

future work

Preliminary Results



Conclusion

- We are in the process of developing and validating an **OSP-based fluorescence microscopy approach** capable of directly imaging O₂ availability inside individual organic aerosol proxy particles, addressing the need for direct measurements.
- The OSP signal responds **sensitively, reversibly, and quantitatively** to O₂ across bulk citric acid solutions and single-particle systems.
- The **environmental flow cell setup** enables systematic control of O₂ concentration, relative humidity, and temperature, providing the experimental foundation to probe phase-state-dependent O₂ diffusion.
- Next steps:** Demonstration of reversible switching of the fluorescent signal in individual particles of different OA types and at different phase states (liquid, semi-solid, solid).

Acknowledgement:

This work is funded by the Novo Nordisk Foundation through grant agreement no. NNF24OC0093745. We thank Jeppe Dalgaard for machining parts of this setup and Klaus Koren for helpful discussions. This work is supported by the Danish National Research Foundation (DNRF172) through the Center of Excellence for Chemistry of Clouds.

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